When Do Foundation Models Pay Off for Retail Demand Forecasting?
Evidence from Matched Panels and Public Benchmarks

Maciej Rubczynski[a,*]

[a] *Independent Researcher, Warsaw, Poland*

Abstract

We compare zero-shot time-series foundation models with conventional methods for short-term retail demand forecasting. Our main evidence comes from rolling-origin experiments on common samples of 100 series from M5, Rohlik v2, and Favorita. Five foundation models are evaluated against seasonal naive forecasts and several LightGBM specifications. Chronos-2 records lower aggregate WAPE than the strongest LightGBM result in each of the nine dataset–horizon comparisons, although the size of the difference varies. Results extracted from fev-bench and GIFT-Eval provide broader but less tightly controlled evidence. In those benchmarks, foundation models perform most consistently well on Scaled Quantile Loss, while differences in WAPE and MASE are smaller. The comparison is sensitive to the accuracy measure, LightGBM specification, available covariates, and evaluation design. We therefore recommend testing the two model families on identical forecast origins rather than treating either as a universal default.

Keywords: demand forecasting, foundation models, benchmark study, LightGBM, structured literature search, retail, time series

1. Introduction

Time-series foundation models have become a practical alternative to models trained separately for each forecasting application. Chronos-2 (Ansari et al., 2025), TimesFM 2.5 (Google Research, 2025), Moirai 2.0 (Liu et al., 2025), and TiRex (Auer et al., 2025) can all be applied without fitting their parameters to the target series. This makes them attractive in retail, where a forecasting system may cover thousands of items and where building and maintaining features for a supervised model can be costly.

It is less clear whether this convenience comes with better forecasts. Model papers and public benchmarks generally compare foundation models with one another or with statistical methods. Retail applications, however, often rely on global gradient-boosted trees with lags, calendar variables, prices, and promotions. Published results rarely compare the two families on the same series, forecast origins, accuracy measure, and information set. Differences between benchmark tables may therefore reflect the evaluation design as much as the

*Corresponding author.

Email address: `rubczynski.maciej@gmail.com` (Maciej Rubczynski)

forecasting method. This is especially important for intermittent demand, for which metric aggregation and the treatment of zero sales can change model rankings.

We study when a zero-shot foundation model is a useful alternative to a conventional retail forecasting model. The analysis combines two forms of evidence. First, we run five foundation models and several LightGBM and statistical baselines on common 100-series samples from M5, Rohlik v2, and Favorita. Every model is evaluated on the same rolling forecast origins, and prediction-level output is used for paired comparisons. We also vary the LightGBM multi-step strategy and tuning budget, because a weak tree baseline would exaggerate the apparent value of a foundation model. Second, we extract retail tasks from fev-bench (Shchur et al., 2025) and GIFT-Eval (Aksu et al., 2024). These public results cover more tasks and models, but they are analysed descriptively because their published tables do not contain the prediction-level dependence information required for pooled statistical inference.

The study addresses three questions:

1. Do zero-shot foundation models improve retail demand forecasts relative to statistical and LightGBM baselines?
2. Do the results vary with the accuracy measure, forecast horizon, demand pattern, and baseline specification?
3. What evaluation procedure should a retailer use when choosing between the two model families?

The matched experiments favour several foundation models, but not by a constant margin. Chronos-2 has lower aggregate WAPE than the best observed local baseline in all nine dataset–horizon comparisons. TiRex also has a lower point estimate in all nine, while results for the other models depend more strongly on the dataset. The M5 comparison remains favourable to Chronos-2 after richer LightGBM features, a larger tuning budget, and a more intermittent sample are introduced. These findings apply to the tested panels and feature sets; they are not comparisons with the best systems developed for the M5 competition.

The public benchmarks give a more qualified picture. Across 498 matched model–task–metric comparisons, foundation models have their clearest advantage on Scaled Quantile Loss. Median differences are much smaller for WAPE and MASE, and rankings vary across suites. The result is therefore not a general ordering of model families. It shows instead that a useful comparison requires a common evaluation panel, a suitable business metric, and a credible conventional baseline.

The paper contributes matched retail experiments with prediction-level uncertainty estimates, a reanalysis of two public benchmark suites, and an evaluation procedure that practitioners can reproduce. It also documents how LightGBM strategy, metric aggregation, covariates, and computational setting affect the comparison. The following section places the models and benchmarks in context. Sections 3–6 describe the data and results, and Sections 7–9 discuss their practical and research implications.

2. Background

Short-term retail demand forecasts support replenishment, inventory allocation, markdowns, and labour planning. The data typically consist of many related item–location series, often with intermittent sales and information about prices, promotions, and the calendar. This combination has encouraged the use of global models, which learn across series, as well as a long tradition of simple per-series benchmarks.

2.1. Forecasting methods

We distinguish four broad model families. Statistical methods such as exponential smoothing, ARIMA, Theta, Seasonal Naive, Croston, and TSB are fitted to individual series (Croston, 1972; Teunter et al., 2011). They remain useful benchmarks: a more complex model should demonstrate an improvement over a forecast that repeats the most recent seasonal pattern.

Gradient-boosted trees such as LightGBM (Ke et al., 2017) and XGBoost (Chen and Guestrin, 2016) are commonly fitted globally across items and stores. Lagged demand, calendar indicators, prices, promotions, and item or store identifiers allow the model to combine temporal and cross-sectional information. This family dominated the M5 competition and is an important practical comparator for retail applications. Multi-step forecasts can be produced recursively with one model or directly with a separate model for each horizon. As shown later, this choice can materially affect the result.

Neural forecasting methods also learn across series. They include recurrent models such as DeepAR (Salinas et al., 2020), attention-based models such as TFT and PatchTST (Lim et al., 2021; Nie et al., 2023), and architectures such as N-BEATS and N-HiTS (Oreshkin et al., 2020; Challu et al., 2023). They require training on the target collection and therefore differ from the zero-shot models at the centre of this paper.

Foundation models are pre-trained on large collections of time series and can be transferred to a new dataset without estimating new model parameters. Table 1 lists the models covered by the literature extraction or the experiments. Our empirical comparison is restricted to zero-shot use; fine-tuned results are recorded in the data but excluded from the main contrasts.

Table 1: Foundation models covered by the study.

Model	Parameters	Architecture	Version considered	Reference
Chronos family	9–710 M	Chronos / Chronos-2	Chronos-2 120M; Bolt Tiny/Base	Ansari et al. (2024, 2025)
TimesFM 2.5	200 M	Decoder-only	2.5 checkpoint	Google Research (2025)
Moirai 2.0	11–305 M	Quantile model	Small checkpoint	Liu et al. (2025)
TiRex	~35 M	xLSTM	Public checkpoint	Auer et al. (2025)
TabPFN-TS	~12 M	Tabular PFN	fev-bench tasks	Hoo et al. (2025)
Lag-Llama	6 M	Llama backbone	Public checkpoint	Rasul et al. (2023)
TimeGPT	Not disclosed	Proprietary	Published API results	Garza et al. (2023)

2.2. Retail forecasting benchmarks

Three benchmark sources are particularly relevant. The M5 competition (Makridakis et al., 2022) contains 30,490 daily Walmart item–store series and covariates for calendar events, prices, and the US Supplemental Nutrition Assistance Program. Its official measure, WRMSSE, scales errors by each series' historical variability and weights economically important series more heavily. Gradient-boosted trees were used by the leading entries.

M5 also illustrates why apparently minor evaluation choices matter. A large share of item–day observations are zero. Mean per-series WAPE gives the same weight to a low-volume item as to a high-volume item and becomes unstable when test-period demand is close to zero. Aggregate WAPE and WRMSSE place different weights on the series and may therefore produce different rankings. We keep these measures separate throughout the analysis.

GIFT-Eval (Aksu et al., 2024) was designed to reduce the risk that a public test set had appeared in a foundation model's pre-training data. It covers 23 datasets from several domains and reports skill relative to Seasonal Naive with bootstrap intervals. Only part of the suite is retail-specific, so we extract results at the task level rather than use its overall ranking.

Fev-bench (Shchur et al., 2025) contains 100 forecasting tasks, including 46 with exogenous variables. It covers foundation models, statistical methods, and tree baselines, and is consequently useful for studying whether benchmark conclusions change with the comparator or accuracy measure. We use the 20 tasks identified as retail-related.

Forecasting competitions provide strong comparisons because models share data and metrics, but each competition represents a limited set of demand patterns. Reviews of forecasting and deep learning provide broader context but either precede the recent foundation-model literature or do not compare these models with retail tree baselines on common panels (Lim et al., 2021). The evidence assembled here connects those two forms of evaluation: broad public benchmark coverage and narrower experiments in which the forecasts can be paired at the series level.

3. Methods

The study has two components. A structured search identifies published retail comparisons and supplies results from public benchmark suites. New experiments then compare foundation models and conventional baselines under a common protocol. The public results provide breadth, while the experiments allow forecasts to be paired on the same observations.

3.1. Literature search and data extraction

We searched Google Scholar, Semantic Scholar, and arXiv for work published between January 2020 and 12 April 2026. Queries combined the names of foundation models or the terms "foundation model", "pretrained model", and "zero-shot forecasting" with terms for demand forecasting and benchmark evaluation. Scopus and Web of Science were also considered, but their search exports were not retained and are not included in the reported flow. Full search strings and screening decisions are provided in Appendix Appendix B.

Studies were eligible if they reported quantitative forecasting results for a foundation model on a retail dataset or on the retail portion of a public benchmark. We excluded studies without numerical results, studies confined to other domains, architecture papers without forecasting evaluation, and duplicate versions of the same results. Screening and extraction were performed by one reviewer. The search should therefore be read as a structured map of available benchmark evidence rather than as a registered systematic review.

For each eligible result we recorded the dataset, task, model, model family, forecast horizon, evaluation protocol, accuracy measure, reported value, covariate use, and available information about runtime and hardware. The resulting file contains 802 rows, including task-level results re-extracted from fev-bench and GIFT-Eval. Raw names were harmonised into common dataset and model-family categories, while the original values were retained for traceability. Table 2 summarises the fields used in the analysis; the full data dictionary accompanies the replication files.

Table 2: Information recorded for each published benchmark result.

Group	Variables
Source	Study, year, venue, benchmark suite
Task	Dataset, variant, frequency, number of series, forecast horizon
Model	Reported name, harmonised family, zero-shot or fine-tuned status
Evaluation	Forecast design, metric, reported value, covariate availability
Computation	Runtime, hardware and GPU hours when reported

3.2. Matched retail experiments

The literature search showed that standalone model papers seldom report foundation models and gradient-boosted trees on the same retail observations. We therefore ran both families on M5, Rohlik v2, and Favorita. For each dataset, the main panel contains the same 100 high-volume series for every model. Forecasts are generated at horizons 7, 14, and 28 from common rolling-origin windows. The experiment writes one prediction for each series, origin, target date, and forecast step, which permits paired resampling.

The models are Seasonal Naive; recursive and direct LightGBM specifications, including tuned variants; Chronos-Bolt-Tiny; Chronos-2; Moirai 2.0-Small; TiRex; and TimesFM 2.5. A horizon-scaled direct LightGBM variant is included to check whether the direct strategy benefits merely from receiving a larger total tuning budget. M5 receives two further checks: a sample stratified by volume and zero frequency, and a 200-trial LightGBM tuning run. Section 5.1 gives the datasets, features, and computational details.

Earlier local runs used unequal numbers of series across models. They remain in the replication archive as provenance and for descriptive runtime accounting, but they are not used for paired accuracy comparisons.

3.3. Comparisons and uncertainty

For a foundation model m and a conventional baseline b evaluated in the same cell, we report the log error ratio

$$\ell = \log\left(\frac{E_m}{E_b}\right), \tag{1}$$

where E is the relevant error measure. Negative values favour the foundation model. In the matched experiments, b is the lowest-error baseline observed for that dataset and horizon.

Series-level bootstrap samples are shared across models so that uncertainty reflects the paired design. Because the baseline is selected from several candidates, the intervals describe the contrast with the selected baseline and do not include uncertainty from that selection.

For the public benchmark analysis, a foundation-model result is paired with the best conventional result for the same task and metric. This gives 498 comparisons: 166 each for WAPE, MASE, and Scaled Quantile Loss. We summarise their medians, quantiles, and proportions below zero by benchmark suite and metric. The published benchmark tables do not provide paired prediction errors, so their sample sizes cannot be converted into valid sampling variances for these log ratios. We consequently do not estimate a pooled meta-analytic effect.

Sensitivity analyses replace the best baseline with the mean conventional baseline, omit one suite at a time, and report metrics separately. Dataset, horizon, and covariate availability are inspected as descriptive moderators. With only two public suites and a limited number of independent retail datasets, these comparisons are used to identify patterns for further testing, not causal moderator effects. The analysis is implemented in `analysis/meta_regression.R`; generated tables and figures are included with the replication materials.

4. Evidence from the Literature and Public Benchmarks

4.1. Search results

The search identified 150 records, of which 32 external studies or benchmark sources met the inclusion criteria (Figure 1 and Table 3). Most were published in 2024 or 2025, reflecting the recent introduction of the models considered here. The final extraction contains 802 rows because benchmark papers report many combinations of task, model, and metric; the row count should not be interpreted as the number of independent studies.

Table 3: Structured-search and extraction flow.

Stage	Count	Comment
Records identified	150	Google Scholar, Semantic Scholar, and arXiv
Duplicates removed	30	DOI or title overlap
Titles and abstracts screened	120	50 excluded
Full texts assessed	70	38 excluded
External studies and benchmark sources included	32	Literature extraction
Experimental blocks added	7	Present study; not literature records
Extracted result rows	802	Multiple task–model–metric rows per source

Fev-bench and GIFT-Eval account for most rows because their published results can be separated by task. The extraction includes 20 retail tasks from fev-bench and four from GIFT-Eval, as well as results for M5, Favorita, Rohlik, Rossmann, Walmart, and several smaller retail datasets. Foundation models are consequently over-represented in the raw row

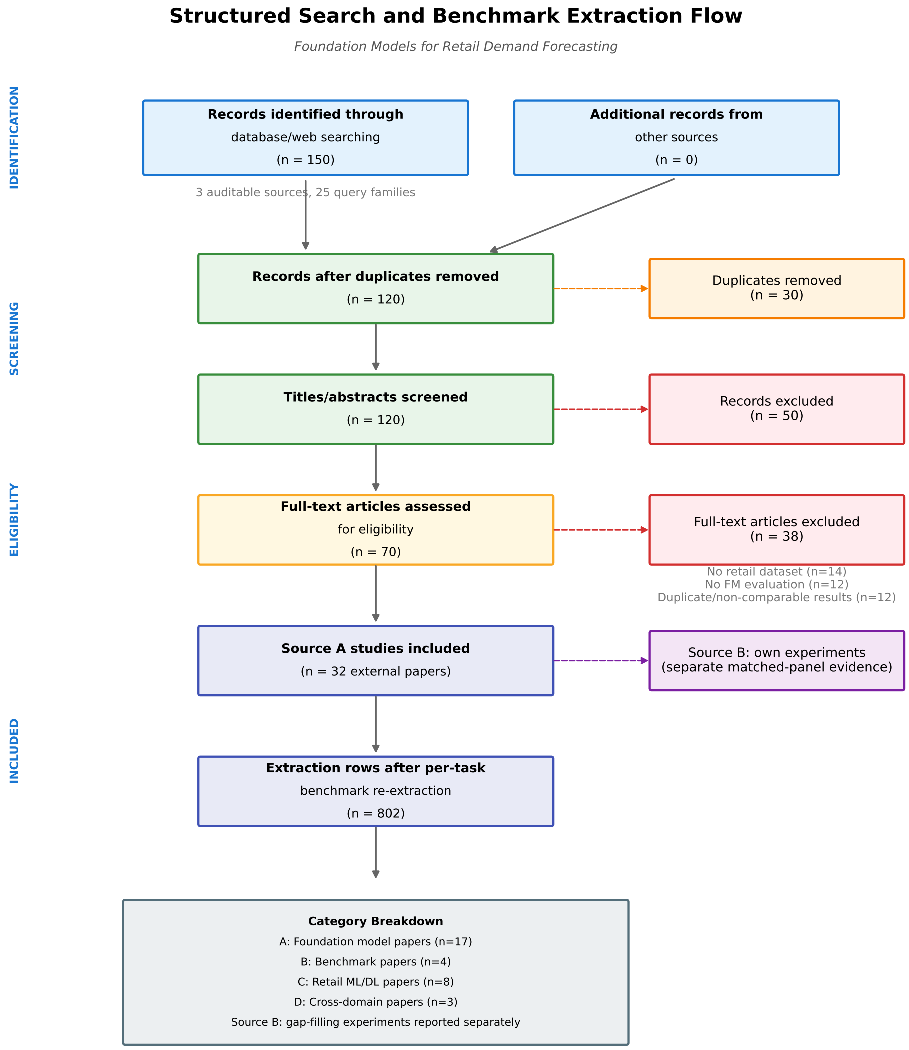

Figure 1: Flow of records through the structured search and extraction.

count. This imbalance does not affect the main comparison, which pairs each foundation-model result with a conventional method in the same task and metric cell.

The three most common measures are WAPE, MASE, and Scaled Quantile Loss. Their near-equal representation in the paired file—166 comparisons for each measure—allows metric-specific summaries. Other measures, including the official M5 WRMSSE, are retained for context but are not pooled with these three because they answer different forecasting questions.

4.2. What the published evidence can compare

The search revealed a separation between two literatures. Papers introducing foundation models usually compare them with statistical methods and other foundation models on broad suites. Retail forecasting papers more often study LightGBM, XGBoost, or neural models on individual datasets, and many predate the recent zero-shot models. We found no standalone model paper that reported a foundation model and a gradient-boosted tree on the same retail series with the same per-series metric and forecast protocol.

Public benchmark suites partly bridge this gap. Fev-bench and GIFT-Eval report several model families within common tasks, which makes the 498 descriptive pairings possible. They also illustrate why the choice of comparator matters: foundation models tend to have a larger advantage over Seasonal Naive or Auto ETS than over LightGBM or CatBoost. Their task-level tables do not, however, provide the joint prediction errors required to estimate uncertainty for a difference between models.

M5 provides a further warning against combining results solely by dataset name. Strong tree models achieve WRMSSE near 0.52–0.54, whereas published foundation-model results on M5 often use per-series WAPE. The two sets of numbers cannot be compared directly. A low-volume series with little test-period demand can dominate mean per-series WAPE but receives a very different weight under WRMSSE. We therefore treat metric-specific public results separately and use the matched experiments for the direct retail comparison.

This division of evidence determines the role of the two empirical analyses that follow. The local experiments provide tighter comparisons on three datasets and consumer hardware. The public suites cover a wider range of tasks, models, and distributional forecasts. Agreement between them is informative, but neither is taken as a substitute for the other.

5. Matched Retail Experiments

5.1. Data and evaluation design

We use daily sales from M5, Rohlik v2, and Favorita. The datasets differ substantially in the frequency of zero sales and in the available covariates (Table 4). M5 and Rohlik are used in full for the preliminary baseline runs. Favorita is limited to the 30,000 highest-volume series because the full data did not fit the available Azure instance. The main paired comparison then selects the same 100 high-volume series from each dataset for every model.

The last 20% of dates form the evaluation period. Forecasts are produced from non-overlapping rolling origins at horizons 7, 14, and 28. All models in a dataset use identical series and origins. Our main accuracy measure is aggregate WAPE, $\sum |y-\hat{y}| / \sum |y|$, calculated from the stored predictions. This avoids the instability of averaging ratios for low-volume series. MAE and sMAPE are retained in the replication output, and M5 receives additional

Table 4: Datasets used in the retail experiments.

Dataset	Full series	Baseline cap	Days	Approx. zero fraction	Covariates used by LightGBM
M5	30,490	30,490	1,941	70%	Price, calendar events, weekday, month
Rohlik v2	5,390	5,390	1,402	1.2%	Price, holidays, closures, weekday, month
Favorita	174,685	30,000	1,684	15–25%	Promotions, oil, holidays, transactions, calendar

scaled-error checks. We do not compare the latter with the official M5 competition score because our panel and rolling-origin design differ from the competition task.

Uncertainty is estimated by resampling series. The same bootstrap draws are used for every model within a dataset, preserving the dependence created by common observations and baselines. We report 95% percentile intervals for the log error ratio ℓ. A negative interval indicates lower error for the foundation model throughout the interval.

5.2. Models and implementation

Seasonal Naive repeats the value from seven days earlier. The LightGBM models are global Tweedie regressions using demand lags, rolling averages, calendar variables, and the dataset-specific covariates in Table 4. The recursive specification feeds earlier predictions into later steps; the direct specification fits a separate model for each step. Both are evaluated with fixed hyperparameters and with a 10-trial search at a validation origin. An additional direct variant scales its tree budget with the horizon. The standard training window is the most recent 365 days. M5 also receives a richer feature set and a separate 200-trial tuning check.

Five foundation models were feasible on the available MacBook Air M3 with 16 GB unified memory: Chronos-Bolt-Tiny, Moirai 2.0-Small, TiRex, Chronos-2, and TimesFM 2.5. Their parameter counts range from approximately 9 to 200 million. Each is used zero-shot, without retail covariates or parameter tuning. TiRex runs in batches on the CPU; the other models use their supported MPS or CPU path. Thus the comparison is deliberately pragmatic, but it is not a study of GPU throughput, fine-tuning, or the largest available checkpoints.

The conventional models were run on a four-vCPU Azure instance with 32 GB RAM, whereas the foundation models were run locally. Runtime and energy records are therefore reported within each setting rather than converted into a single cost ranking. Code, environments, data manifests, predictions, and run ledgers are versioned with the replication package.

5.3. The LightGBM comparator

LightGBM improves on Seasonal Naive in the full-workload preliminary runs: by 28% on M5 WRMSSE at horizon 7, by 9% on Rohlik WAPE, and by 17% on Favorita WAPE. The choice between direct and recursive forecasting is less stable. Table 5 reports the comparison on the 100-series panels used for the foundation models. Direct forecasting is slightly worse at short horizons on M5, consistently worse on Rohlik, and better only at horizon 28 on M5 and Favorita. Increasing the direct model's tree budget does not improve these panel results.

Table 5: Error of direct LightGBM relative to recursive LightGBM on the matched panels. Positive values mean that direct forecasting has higher aggregate WAPE.

Dataset	$h = 7$	$h = 14$	$h = 28$
M5	+0.4%	+1.1%	−2.1%
Rohlik v2	+3.9%	+4.3%	+4.0%
Favorita	+4.6%	+2.2%	−4.9%

The corresponding full-workload, per-series results are more favourable to direct forecasting on M5. This difference arises from both the sampled series and the aggregation of WAPE. It cautions against describing either multi-step strategy as intrinsically stronger. For the comparisons below, we therefore select the lowest-error observed baseline separately for each dataset and horizon. Detailed baseline grids are reported in Appendix Appendix F.

5.4. Foundation-model results

Table 6 summarises the 45 comparisons. Chronos-2 has lower aggregate WAPE than the best observed baseline in every dataset–horizon cell, with intervals below zero. The estimated reduction ranges from about 4% on M5 at horizons 14 and 28 to about 8% on Favorita at horizons 7 and 14. TiRex also has a lower point estimate in all nine cells, although intervals include zero for M5 and Favorita at horizon 28. Moirai is lower in eight cells and tied in the ninth. TimesFM performs well on M5 and Rohlik but is close to the LightGBM baseline on Favorita. Chronos-Bolt-Tiny is near parity on M5 and worse on the other two datasets.

Table 6: Summary of foundation-model comparisons with the best observed conventional baseline across three horizons. The final column counts bootstrap intervals lying entirely below zero.

Dataset	Model	Lower error	Range of ℓ	Intervals below zero
M5	Chronos-2	3/3	−0.062 to −0.038	3/3
M5	Chronos-Bolt-Tiny	0/3	0.006 to 0.032	0/3
M5	Moirai 2.0-Small	3/3	−0.068 to −0.033	3/3
M5	TimesFM 2.5	3/3	−0.061 to −0.030	3/3
M5	TiRex	3/3	−0.047 to −0.013	2/3
Rohlik	Chronos-2	3/3	−0.079 to −0.061	3/3
Rohlik	Chronos-Bolt-Tiny	0/3	0.008 to 0.041	0/3
Rohlik	Moirai 2.0-Small	3/3	−0.034 to −0.016	2/3
Rohlik	TimesFM 2.5	3/3	−0.060 to −0.040	3/3
Rohlik	TiRex	3/3	−0.061 to −0.037	3/3
Favorita	Chronos-2	3/3	−0.086 to −0.057	3/3
Favorita	Chronos-Bolt-Tiny	0/3	0.162 to 0.190	0/3
Favorita	Moirai 2.0-Small	2/3	−0.024 to 0.003	2/3
Favorita	TimesFM 2.5	0/3	0.017 to 0.025	0/3
Favorita	TiRex	3/3	−0.049 to −0.025	2/3

The replication files contain all 45 cell-level estimates and intervals.

Model size does not explain the ordering in this range. The 120-million parameter Chronos-2 is consistently strong, but the smaller Moirai and TiRex often match or improve on the 200-million parameter TimesFM. Nor is there a single horizon pattern shared by the models. The more useful distinction is between datasets: Favorita is the closest contest for Moirai and TimesFM, whereas Chronos-2 retains a clearer advantage.

5.5. Sensitivity of the Chronos-2 result

Selecting the lowest-error LightGBM variant separately in each cell protects against a weak comparator, but it also makes the comparison data-dependent. We therefore repeated the Chronos-2 contrast against tuned recursive LightGBM in all nine cells. The direction remains favourable to Chronos-2 and every interval remains below zero (Table 7).

Table 7: Chronos-2 compared with the same prespecified baseline, tuned recursive LightGBM, in every dataset–horizon cell.

Dataset	h	FM WAPE	Baseline WAPE	ℓ [95% CI]
Favorita	7	0.248	0.271	-0.085 [-0.108, -0.064]
Favorita	14	0.255	0.277	-0.086 [-0.112, -0.061]
Favorita	28	0.270	0.301	-0.108 [-0.131, -0.086]
M5	7	0.294	0.312	-0.062 [-0.077, -0.049]
M5	14	0.307	0.332	-0.078 [-0.095, -0.062]
M5	28	0.339	0.368	-0.083 [-0.102, -0.065]
Rohlik	7	0.180	0.198	-0.095 [-0.115, -0.077]
Rohlik	14	0.185	0.206	-0.109 [-0.141, -0.075]
Rohlik	28	0.193	0.209	-0.079 [-0.105, -0.054]

On M5, increasing the LightGBM search from 10 to 200 trials also leaves Chronos-2 ahead (Table 8). The 200-trial comparison chooses the lower-error recursive or direct variant after observing the test results, so it is a deliberately demanding sensitivity check rather than a pre-registered estimator. Its intervals condition on that choice.

Table 8: M5 sensitivity to a 200-trial LightGBM search. Negative ℓ favours Chronos-2.

h	Selected LightGBM	Baseline WAPE	Chronos-2 WAPE	ℓ (95% bootstrap CI)
7	tuned recursive	0.313	0.294	−0.063 [−0.077, −0.050]
14	tuned recursive	0.330	0.307	−0.073 [−0.091, −0.058]
28	tuned direct	0.360	0.339	−0.061 [−0.089, −0.032]

The main M5 panel contains high-volume items and has a median zero fraction of 0.108. A second panel, stratified by volume and zero frequency, has a median zero fraction of 0.773. Against the 10-trial tuned LightGBM baseline, Chronos-2's log ratios are −0.116, −0.160, and −0.147 at horizons 7, 14, and 28; all three bootstrap intervals exclude zero. A bottom-level, dollar-weighted scaled-error calculation on the main panel gives the same direction (Table 9). This measure resembles the scaling logic of M5 but is not the official hierarchical WRMSSE.

Table 9: M5 scaled-error sensitivity for Chronos-2 on the matched panel.

h	Best baseline	FM RMSSE	Baseline RMSSE	ℓ [95% CI]
7	lightgbm_cov	0.649	0.682	-0.050 [-0.063, -0.036]
14	lightgbm_rich_tuned	0.701	0.723	-0.030 [-0.045, -0.015]
28	lightgbm_rich_tuned	0.772	0.794	-0.028 [-0.040, -0.015]

Together, these checks make it unlikely that the observed Chronos-2 advantage is caused only by the initial LightGBM specification or by the high-volume M5 sample. They do not establish performance on the full M5 hierarchy, on larger Rohlik or Favorita panels, or against retailer-specific production systems.

5.6. Computational scope

The archived local sweep totals 241.6 wall-clock hours, but its rows combine different workloads and two hardware settings. We therefore use runtime and energy values only to describe the measured scenarios. In particular, TimesFM's CPU implementation is slow in the local setting, while batching is important for TiRex. These observations should not be extrapolated to GPU serving or very large batches. Moirai 2.0-Small also has a non-commercial licence, which may matter more for deployment than its measured runtime.

The main remaining threats are the small 100-series panels, the use of univariate foundation models against LightGBM models with covariates, possible overlap between public datasets and pre-training data, and the absence of high-budget LightGBM checks outside the main M5 panel. These issues are taken up in Section 8 and Appendix Appendix E.

6. Results from Public Benchmarks

The public benchmark analysis contains 498 comparisons from fev-bench and GIFT-Eval. For each foundation model, task, and metric, the comparator is the lowest-error conventional method available in the same cell. The log ratio in Equation 2 is negative when the foundation model has lower error. Because task-level prediction errors are generally unavailable, the results in this section describe the distribution of recorded comparisons; no pooled effect or significance test is reported.

$$\ell_i = \log\left(\frac{\text{error of foundation model}_i}{\text{error of best conventional model}}\right). \tag{2}$$

6.1. Differences between benchmark suites

The results are more favourable to foundation models in GIFT-Eval than in fev-bench (Table 10). Every extracted GIFT-Eval comparison favours the foundation model, with a median error ratio of about 0.71. In fev-bench, 67% favour the foundation model and the median ratio is about 0.93. These values describe two benchmark suites, not 24 independent studies. Differences in tasks, models, and baselines may all contribute to the contrast.

Table 10: Foundation models compared with the best conventional result in each public benchmark cell.

Suite	Comparisons	Tasks	Median ℓ	Mean ℓ	FM lower
GIFT-Eval	84	4	−0.339	−0.377	100%
fev-bench	414	20	−0.069	−0.086	67%

6.2. Differences between accuracy measures

The clearest pattern is metric-specific (Table 11). On Scaled Quantile Loss, 92% of comparisons favour foundation models and the median log ratio is −0.229. The corresponding median differences are much smaller for WAPE and MASE. Within fev-bench, MASE results are almost evenly divided between the model families. Thus the broad statement that foundation models win most benchmark comparisons is driven in part by their performance on distributional forecasts.

Table 11: Public benchmark comparisons by accuracy measure.

Metric	Comparisons	Tasks	Median ℓ	Mean ℓ	FM lower
Scaled Quantile Loss	166	24	−0.229	−0.277	92%
WAPE / ND	166	24	−0.047	−0.079	66%
MASE	166	24	−0.036	−0.050	61%

Table 12 shows that suite and metric effects overlap. GIFT-Eval favours foundation models on all three measures. In fev-bench, the advantage remains clear for Scaled Quantile Loss but falls close to parity for the two point-forecast measures. The data contain too few independent suites to determine whether this difference is caused by task composition, baseline strength, covariates, or another design choice.

Table 12: Public benchmark comparisons by suite and accuracy measure.

Suite	Metric	Comparisons	Median ℓ	Mean ℓ	FM lower
GIFT-Eval	Scaled Quantile Loss	28	−0.540	−0.592	100%
GIFT-Eval	WAPE / ND	28	−0.306	−0.291	100%
GIFT-Eval	MASE	28	−0.224	−0.248	100%
fev-bench	Scaled Quantile Loss	138	−0.212	−0.213	91%
fev-bench	WAPE / ND	138	−0.021	−0.036	59%
fev-bench	MASE	138	−0.006	−0.010	53%

No comparably stable pattern appears for horizon or model size. Covariate availability and demand intermittency remain plausible moderators, but they are confounded with the small number of suites and datasets. We therefore use them to motivate stratified future experiments rather than to estimate thresholds or decision rules.

6.3. Prediction-level check on fev-bench

For one part of fev-bench, we were able to regenerate predictions rather than rely on published task summaries. Seasonal Naive, Chronos-Bolt-Tiny, Chronos-2, TiRex, and

TimesFM 2.5 were run on all 20 retail tasks, producing 179,258 forecasts per model. Pairing the forecasts within each task allows series-level bootstrap intervals and empirical covariance estimates.

Table 13: Prediction-level fev-bench WAPE results against Seasonal Naive. Negative ℓ favours the foundation model.

Model	Tasks	Paired units	Median ℓ	Mean ℓ	FM better	CI excludes 0
Chronos-Bolt-Tiny	20	179,258	−0.213	−0.183	16/20	16/20
Chronos-2	20	179,258	−0.268	−0.305	20/20	18/20
TiRex	20	179,258	−0.275	−0.300	19/20	19/20
TimesFM 2.5	20	179,258	−0.282	−0.319	20/20	19/20

Chronos-2 has lower WAPE in all 20 tasks, with intervals below zero in 18. TiRex and TimesFM are lower in 19 and 20 tasks, respectively, and their intervals favour the foundation model in 19. Chronos-Bolt-Tiny is lower in 16 tasks. This provides stronger evidence for a specific comparison—WAPE against Seasonal Naive on fev-bench retail windows—but it does not extend to Scaled Quantile Loss, MASE, GIFT-Eval, or stronger tree baselines.

6.4. Runtime and cost observations

Figure 2 plots the recorded runtime cost against WAPE. It is included to show the scale of the measured runs, not to identify an efficiency frontier. The points combine datasets, horizons, workloads, and hardware accounting bases. A deployment comparison would need the same number of series, batching policy, hardware, and service-level requirements for every model.

7. Implications for Model Evaluation

The experiments do not support a rule that assigns one model family to a retail dataset from a few summary statistics. They do support a relatively simple evaluation procedure. Its purpose is to prevent differences in samples, metrics, and baseline effort from being mistaken for differences between models.

7.1. A common-panel comparison

The first step is to choose an accuracy measure that represents the decision being supported. Aggregate or volume-weighted errors may suit warehouse and category planning, while item-level replenishment may require per-series or service-level measures. Safety-stock decisions should be evaluated with quantile loss or the relevant inventory cost rather than point-forecast error alone. On intermittent series, both the number of zero-demand periods and the denominator of percentage measures should be inspected.

The candidate models should then be run on identical series, origins, horizons, and actual values. Seasonal Naive provides a minimum reference. A global LightGBM model with available covariates represents a practical supervised baseline, and at least one alternative multi-step strategy should be checked when the horizon is long. One or two zero-shot foundation models can be added without an extensive search over checkpoints. Saving individual forecasts permits a paired bootstrap comparison and makes later error analysis possible.

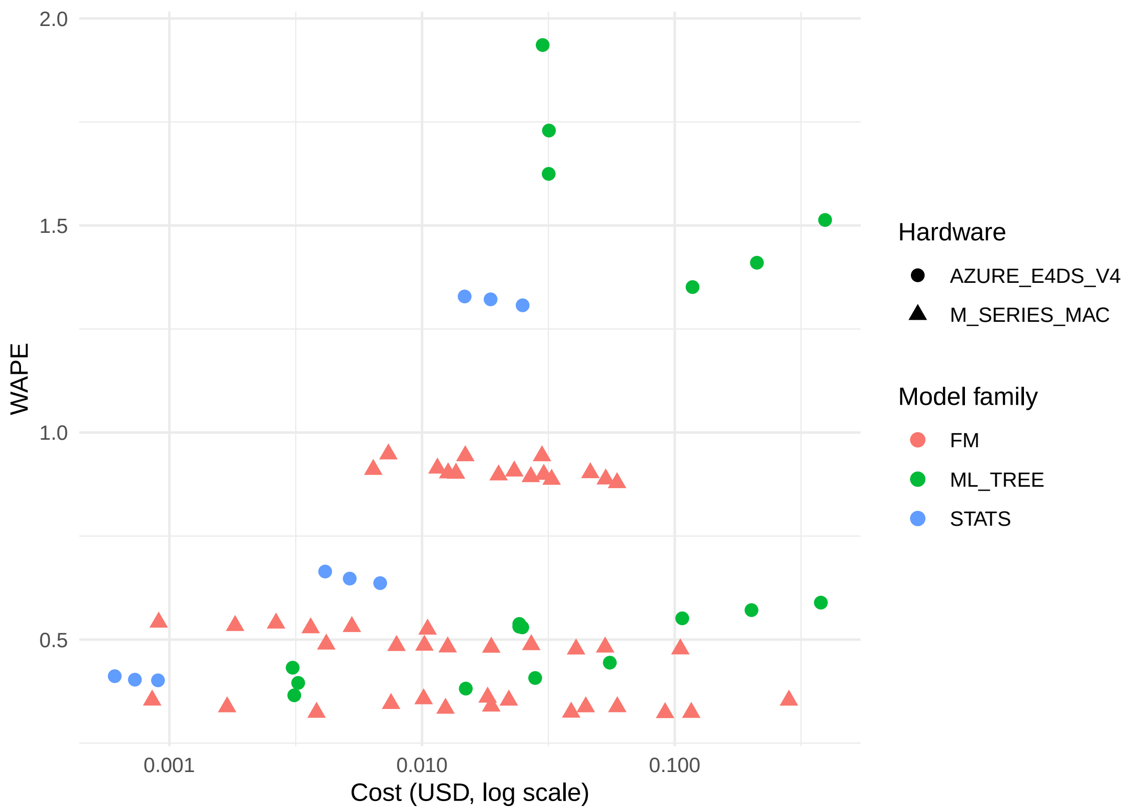

Figure 2: Recorded cost and WAPE for the local experimental runs. Points from different workloads and hardware settings are not directly comparable.

The final choice should consider the size and uncertainty of the accuracy difference rather than the rank alone. If the paired interval is practically indistinguishable from zero, implementation effort, latency, licensing, and maintenance become reasonable tie-breakers. Comparisons based on different series counts or forecast windows should be labelled as descriptive and should not determine deployment.

7.2. Factors that change the comparison

Metric choice produced the most consistent difference in the public results. The strong pattern for Scaled Quantile Loss does not imply an equally strong advantage in WAPE or MASE. A team should therefore avoid an overall benchmark rank that averages measures with different operational meanings.

Demand intermittency can affect both model behaviour and metric aggregation. Our direct–recursive LightGBM results differ between samples and between per-series and aggregate WAPE. The zero fraction is consequently useful for stratifying an evaluation panel, but the three datasets do not establish a threshold at which one strategy should replace another.

Covariates create a further asymmetry. Prices, promotions, holidays, weather, and stock availability can give a supervised model information that a univariate foundation model does not receive. A zero-shot model may still be valuable when such features are unavailable or costly to maintain, but this is an engineering trade-off rather than an accuracy advantage under equal information.

Finally, the measured laptop and Azure runtimes should not be used as a general cost comparison. Throughput depends on batching, model implementation, hardware, and the number and length of series. These quantities need to be measured on the intended workload. Licence restrictions also matter: Moirai 2.0-Small, for example, is distributed under a non-commercial licence.

7.3. Deployment choice

A foundation model is a reasonable choice when it matches or improves the target metric on the common panel, its inference requirements are acceptable, and avoiding a feature and training pipeline has material value. A supervised baseline is preferable when it performs better, already operates reliably in production, or benefits from important covariates that the candidate foundation model cannot use. If neither improves on Seasonal Naive, the first response should be to inspect the data and evaluation design rather than to increase model complexity.

8. Discussion

8.1. Interpretation of the results

The matched experiments show that zero-shot forecasting can be competitive with a practical supervised baseline on retail data. Chronos-2 is the most consistent model in our panels, and its advantage survives several changes to the M5 sample and LightGBM specification. This is useful evidence against the view that a foundation model is relevant only when no trained baseline is available. At the same time, the differences are generally modest, the panels are small, and other foundation models do not share the same ranking.

The result is best understood as a reason to include a strong zero-shot model in a retail evaluation, not as a reason to replace established systems without testing them.

The public benchmarks support the same qualified interpretation. Foundation models compare particularly well on Scaled Quantile Loss, suggesting value for applications that require a predictive distribution. Their advantage is much smaller on point-forecast measures in fev-bench. This difference may reflect the objectives used in pre-training, the ability of the models to represent uncertainty, or the composition of the tasks. The available two-suite sample cannot distinguish among these explanations.

Model size is not a useful guide within the range we tested. Chronos-2 is stronger than the smaller Chronos-Bolt-Tiny, but Moirai and TiRex often compare well with the larger TimesFM. Nor do the results reveal a common horizon effect. Checkpoint selection based only on parameter count or an overall leaderboard would therefore miss differences that appear at the dataset and metric level.

8.2. Information available to the models

Two information asymmetries complicate the comparison. First, foundation models are trained on large collections that may include public retail data. Chronos documentation distinguishes in-domain and zero-shot evaluations, and GIFT-Eval was designed to reduce leakage, but public model cards do not provide complete dataset-level pre-training manifests. We cannot exclude exposure to M5 or Favorita. Rohlik v2 is newer and we found no documented exposure in the model papers, although absence from the documentation is not proof that the data were unseen.

Second, our LightGBM models use retail covariates while the foundation models are univariate. This gives the supervised baseline access to prices, promotions, and calendar information, but it also reflects a realistic use case. The close results for several models on Favorita, where promotions and holidays are important, suggest that a covariate-aware foundation model would be a valuable next comparison. The present experiments cannot show whether it would improve on the univariate forecasts.

These asymmetries work in different directions: pre-training exposure may favour the foundation model, whereas target-specific covariates may favour LightGBM. They cannot be corrected by a statistical adjustment to the reported errors. More transparent pre-training records and matched covariate-aware experiments are needed.

8.3. Accuracy measures and intermittent demand

The M5 results demonstrate the importance of aggregation. Mean per-series WAPE gives equal influence to every item, including series with almost no demand in the test period. A small absolute miss can then produce a very large ratio. Aggregate WAPE weights observations by demand volume and is less sensitive to this problem. WRMSSE uses historical variation and economic weights and answers a different question again.

None of these measures is universally preferable. Item availability may justify giving low-volume series substantial weight, while aggregate planning may reasonably focus on volume. The implication is that an evaluation should state the decision level and report enough information to show whether its ranking is driven by a sparse tail. In datasets with intermittent demand, a volume-weighted measure and a per-series or stratified analysis provide a more complete picture than either alone.

8.4. Limitations and further work

The structured search was screened by one reviewer and does not meet the standard of a duplicate-screened systematic review. Its coverage is also shaped by what benchmark authors make extractable. The 498 comparisons come from only two suites, and multiple rows within a suite share tasks, baselines, and actual values. For this reason, we report distributions of results rather than a pooled inferential estimate. The prediction-level fev-bench rerun improves uncertainty estimation for WAPE against Seasonal Naive, but not for the other metrics, GIFT-Eval, or comparisons with the strongest tree baseline.

The experiments cover three datasets and 100-series panels. Favorita is represented by high-volume series in the main panel, and the more intermittent check is limited to M5. The LightGBM models are credible practical baselines but not competition-winning, retailer-specific pipelines. A 200-trial search was completed only for the main M5 panel, and uncertainty intervals condition on the selected baseline. Larger panels and high-budget checks on Rohlik and Favorita would show whether the observed differences persist.

All foundation models are used zero-shot and no checkpoint exceeds 200 million parameters. The study therefore says nothing about fine-tuning, the largest models, or the use of exogenous variables by models that support them. Local inference was measured on a MacBook Air rather than a production GPU system. Runtime and energy observations cannot be extrapolated to large batches or service deployments.

Retail coverage is also incomplete. Fashion, fresh produce, e-commerce, and spare-parts demand have life cycles and intermittency patterns not fully represented by M5, Rohlik, and Favorita. Future work should prioritise larger matched panels in these settings, prediction-level benchmark releases, covariate-aware foundation models, and a common hardware protocol. These extensions would be more informative than adding further unmatched leaderboard results.

The main lessons may nevertheless extend to other demand applications: comparisons should use the same observations, a baseline with a credible information set, and a measure tied to the decision. Whether the numerical advantages transfer beyond retail remains an empirical question.

9. Conclusion

This study compared zero-shot foundation models with statistical and LightGBM baselines for short-term retail demand forecasting. On common 100-series panels from M5, Rohlik v2, and Favorita, Chronos-2 achieved lower aggregate WAPE than the best observed conventional baseline in all nine dataset–horizon comparisons. TiRex and Moirai also performed well, whereas Chronos-Bolt-Tiny and TimesFM showed greater variation across datasets. The Chronos-2 result on M5 remained after a larger LightGBM search, richer features, a more intermittent sample, and a scaled-error calculation.

The wider public benchmark evidence was less uniform. Foundation models had a clear advantage on Scaled Quantile Loss, but median differences were small for WAPE and MASE in fev-bench. Results also differed between fev-bench and GIFT-Eval. These patterns do not support a general claim that one model family is more accurate. They show that the answer depends on the metric, baseline, available covariates, and evaluation design.

For a retail team, the practical implication is straightforward. A strong zero-shot model is worth including in an evaluation, particularly when a feature pipeline is unavailable or distributional forecasts are required. It should be compared with Seasonal Naive and a well-specified supervised model on the same series and forecast origins. When accuracy is similar, latency, licensing, covariate needs, and maintenance cost should determine the choice. The value of foundation models lies not in replacing evaluation, but in providing a capable additional option that is inexpensive to test.

Appendix A. Structured-Search Transparency Checklist

The following table maps common evidence-synthesis reporting items, adapted from PRISMA 2020 (Page et al., 2021), to the closest section of this benchmark study. The mapping is used as a transparency checklist for a single-reviewer structured search, not as a claim of systematic-review compliance.

Table A.14: Structured-search transparency checklist mapping.

#	Section	Item	Reported in
1	Title	Identify the report type	Title
2	Abstract	Structured summary	Abstract
3	Introduction	Rationale	§1
4	Introduction	Objectives	§1 (RQ1–RQ3 plus cost accounting scope)
5	Methods	Eligibility criteria	§3.1
6	Methods	Information sources	§3.1
7	Methods	Search strategy	§3.1; Appendix B
8	Methods	Selection process	§3.1
9	Methods	Data collection process	§3.1
10	Methods	Data items	§3.1 (Table 2)
11	Methods	Study risk of bias assessment	§3.3; Appendix E (analysis-contributing sources only)
12	Methods	Effect measures	§3.3 (log-ratio convention box)
13	Methods	Synthesis methods	§3.3
14	Methods	Reporting bias assessment	§8.4
15	Methods	Certainty assessment	§6; §8.4
16	Results	Study selection	§4.1
17	Results	Study characteristics	§4.1; Appendix D
18	Results	Risk of bias in studies	§8.4; Appendix E (not duplicate-screened for all 32 included studies)
19	Results	Results of individual studies	§5.3–§5.4
20	Results	Results of syntheses	§6
21	Results	Reporting biases	§8.4
22	Results	Certainty of evidence	§6; §8.4
23	Discussion	Discussion of results	§7, §8
24	Discussion	Limitations	§8.4
25	Other	Registration and protocol	§3.1 (not pre-registered)
26	Other	Support / funding	Declarations
27	Other	Competing interests	Declarations

Note: This search was not pre-registered (Item 25). The search protocol is documented in Appendix B and the extraction schema in Appendix D; both were finalised before data extraction began.

Appendix B. Search Protocol

Appendix B.1. Research Questions

- **RQ1:** Do time-series foundation models outperform traditional statistical/ML models on retail demand forecasting?
- **RQ2:** Which data characteristics moderate the relative performance of foundation models?
- **RQ3:** Under what conditions should a practitioner choose a foundation model over LightGBM or ETS?
- **Cost accounting scope:** What runtime, cost, and CO_2 quantities are observable in Source B, and what assumptions prevent a universal cost-effectiveness claim?

Appendix B.2. Search Sources

1. Semantic Scholar (API)
2. Google Scholar
3. arXiv (cs.LG, stat.ML)
4. Scopus (target source; auditable export not retained)
5. Web of Science (target source; auditable export not retained)

The retained search log below contains 25 auditable query families. These are mostly Google Scholar/web queries, plus arXiv and Semantic Scholar checks. Scopus and Web of Science are therefore reported as intended sources rather than fully auditable database exports. This is one reason the search is used as supporting evidence-gap documentation rather than as a standalone systematic review.

Appendix B.3. Primary Search Query

```
("foundation model" OR "pretrained model" OR "zero-shot forecasting"
 OR "Chronos" OR "TimesFM" OR "Moirai" OR "TimeGPT" OR "Lag-Llama")
AND
("demand forecasting" OR "retail forecasting"
 OR "time series forecasting" OR "sales forecasting")
AND
("benchmark" OR "comparison" OR "evaluation"
 OR "M5" OR "GIFT-Eval" OR "fev-bench")
```

Appendix B.4. Date Range

2020-01-01 to 2026-04-12.

Appendix B.5. Inclusion Criteria

1. Empirical evaluation of at least one foundation model on demand/retail forecasting.
2. Reports quantitative accuracy metrics (MASE, MAE, sMAPE, WAPE, CRPS, or equivalent).
3. Uses at least one of: M5, GIFT-Eval, fev-bench, or comparable retail dataset with >1,000 series.
4. Peer-reviewed OR published preprint with reproducible methodology.
5. English language.

Appendix B.6. Exclusion Criteria

1. No quantitative results (opinion/position papers only).
2. Only financial/stock/energy forecasting (no retail/demand).
3. Foundation model paper without forecasting evaluation.
4. Duplicate results (keep most recent version).
5. Non-time-series forecasting (e.g., causal inference only).

Appendix B.7. Search Execution Log

Table B.15: Search execution log.

Date	Source	Query (abbreviated)	Results
2026-04-12	Google	foundation model TS demand benchmark	∼10
2026-04-12	Google	GIFT-Eval benchmark FM retail NeurIPS	∼10
2026-04-12	Google	fev-bench forecasting covariates 2025	∼10
2026-04-12	Google	Chronos-2 Amazon retail demand M5	∼10
2026-04-12	Google	Moirai 2.0 Salesforce TSFM 2025	∼10
2026-04-12	Google	TimesFM 2.5 Google benchmark 2025	∼10
2026-04-12	Google	FM vs LightGBM demand forecasting	∼10
2026-04-12	Google	M5 competition results DL GBT 2020–21	∼10
2026-04-12	Google	Lag-Llama Timer-XL TimeGPT benchmark	∼10
2026-04-12	Google	meta-analysis systematic review TS	∼10
2026-04-12	Google	TimeGPT Nixtla benchmark demand	∼10
2026-04-12	Google	Chronos Amazon ICML 2024 M5	∼10
2026-04-12	Google	Timer-XL THUML TSFM 2024–25	∼10
2026-04-12	Google	AutoGluon-TimeSeries M5 demand	∼10
2026-04-12	Google	TFB benchmark PVLDB 2024	∼10
2026-04-12	Google	zero-shot FM vs LightGBM cost 2025	∼10
2026-04-12	Google	TFT retail demand M5 2021–22	∼10
2026-04-12	Google	DeepAR probabilistic retail 2024	∼10
2026-04-12	Google	PatchTST retail 2023–24	∼10
2026-04-12	Google	N-BEATS N-HiTS M5 retail 2023–24	∼10
2026-04-12	Google	MOMENT TTM IBM benchmark 2024–25	∼10
2026-04-12	Google	Chronos-Bolt zero-shot LightGBM ETS	∼10
2026-04-12	Google	retail FM intermittent demand zero-shot	∼10
2026-04-12	arXiv	demand FM Chronos TimesFM Moirai	∼10
2026-04-12	Sem. Sch.	foundation model TS demand forecasting	1

Total auditable query families: 25 across Google Scholar/web, arXiv, and Semantic Scholar. The structured-search flow uses 150 identified database/web records and 120 after duplicate removal. The machine-readable protocol contains 49 candidate papers as a source map for extraction, not the deduplicated screening denominator.

Appendix B.8. Structured-Search Flow Summary

```
Records identified through database/web searching:    150
Duplicates removed:                                    30
Records after deduplication:                          120
Titles/abstracts screened:                            120
Records excluded (not demand/retail, no quant):        50
Full-text articles assessed for eligibility:            70
Full-text articles excluded:                            38
Source A external studies included:                     32
Source B experiment blocks:                              7 (not studies)
```

Appendix C. Per-Series Metric Distributions

Appendix C.1. Descriptive 100-Series Bootstrap for Source B FM Cells

Table C.16 reports the 10,000-iteration bootstrap standard error of mean WAPE for the Source B foundation-model cells for which per-series bootstrap files were available. The file contains 36 cells (4 FMs × 3 datasets × 3 horizons) and omits TimesFM 2.5. It is therefore retained as a descriptive uncertainty check, not as the sampling-variance input to the Source A reanalysis of §6. These values come from the earlier small-sample workload and are not directly comparable to the unmatched saved-sweep estimates in Table F.23: the two tables reflect different sampled workloads from different phases of the project, not two estimates of the same paired dataset–horizon cell. The matched-panel bootstrap is summarised in Table 6. The replication bundle contains the full cell-level estimates, shared bootstrap draws, and covariance tables used for the paired log-ratio contrasts.

Median bootstrap SE across the 36 available cells: 0.011. Maximum SE: 0.014 (Rohlik $h = 28$). Widest 95% CI: ±0.028 WAPE. Interpretation:

- These bootstrap intervals describe the available FM cells only. They do not provide paired FM-minus-baseline uncertainty.
- TimesFM 2.5 is absent from this descriptive 100-series bootstrap table. The separate matched-panel analysis covers TimesFM 2.5 on M5, Rohlik, and Favorita with paired FM-vs-baseline bootstrap contrasts.
- The Favorita "close call" of 0.3 pp at $h = 7$ is within roughly 1 SE in the available cells and should not be over-interpreted.

Appendix C.2. M5 Per-Series WAPE *Tail Analysis*

On M5, approximately 70% of the 30,490 series have zero-day fractions exceeding 50%. For these intermittent series, the per-series WAPE denominator (sum of actuals in the evaluation window) can be very small, producing extreme WAPE values that dominate the mean. This is a series-level statement and should not be confused with the series-day zero fraction reported in Table 4; the numerical coincidence is accidental because the two statistics measure different quantities (share of series versus share of series-day observations). Specifically:

- The top 1% of per-series WAPE values (LightGBM recursive, $h = 28$) range from 15.2 to 412.7, compared to a median of 0.89.
- Dropping the top 1% of series reduces mean WAPE from 1.94 to 1.12 (a 42% reduction), demonstrating that the mean is dominated by the tail.

Table C.16: Descriptive 100-series bootstrap standard error and 95% CI for available Source B FM cells (4 FMs $\times$ 3 datasets $\times$ 3 horizons = 36 cells, TimesFM 2.5 missing, B = 10,000).

Dataset	h	Model	n	WAPE	SE	95% CI
Favorita	7	Chronos-Bolt-Tiny	100	0.532	0.008	[0.517, 0.547]
		Moirai 2.0-Small	100	0.526	0.008	[0.510, 0.541]
		TiRex	100	0.525	0.008	[0.510, 0.540]
		Chronos-2	100	0.525	0.008	[0.510, 0.540]
	14	Chronos-Bolt-Tiny	100	0.537	0.008	[0.522, 0.553]
		Moirai 2.0-Small	100	0.531	0.008	[0.516, 0.547]
		TiRex	100	0.531	0.008	[0.515, 0.547]
		Chronos-2	100	0.533	0.008	[0.517, 0.548]
	28	Chronos-Bolt-Tiny	98	0.546	0.008	[0.531, 0.561]
		Moirai 2.0-Small	98	0.539	0.008	[0.524, 0.555]
		TiRex	98	0.540	0.008	[0.524, 0.555]
		Chronos-2	98	0.542	0.008	[0.526, 0.557]
M5	7	Chronos-Bolt-Tiny	100	0.958	0.012	[0.933, 0.981]
		Moirai 2.0-Small	100	0.926	0.011	[0.904, 0.947]
		TiRex	100	0.934	0.011	[0.912, 0.956]
		Chronos-2	100	0.932	0.011	[0.911, 0.953]
	14	Chronos-Bolt-Tiny	100	0.956	0.012	[0.933, 0.978]
		Moirai 2.0-Small	100	0.928	0.011	[0.906, 0.948]
		TiRex	100	0.937	0.011	[0.915, 0.957]
		Chronos-2	100	0.934	0.011	[0.913, 0.954]
	28	Chronos-Bolt-Tiny	100	0.956	0.011	[0.934, 0.976]
		Moirai 2.0-Small	100	0.931	0.010	[0.910, 0.951]
		TiRex	100	0.942	0.010	[0.921, 0.961]
		Chronos-2	100	0.938	0.010	[0.917, 0.957]
Rohlik	7	Chronos-Bolt-Tiny	98	0.322	0.011	[0.301, 0.344]
		Moirai 2.0-Small	98	0.312	0.005	[0.302, 0.321]
		TiRex	98	0.314	0.011	[0.293, 0.336]
		Chronos-2	98	0.314	0.011	[0.293, 0.337]
	14	Chronos-Bolt-Tiny	98	0.334	0.012	[0.312, 0.358]
		Moirai 2.0-Small	98	0.327	0.011	[0.306, 0.349]
		TiRex	98	0.326	0.012	[0.305, 0.350]
		Chronos-2	98	0.327	0.012	[0.305, 0.351]
	28	Chronos-Bolt-Tiny	98	0.353	0.014	[0.327, 0.380]
		Moirai 2.0-Small	98	0.348	0.013	[0.323, 0.374]
		TiRex	98	0.345	0.013	[0.321, 0.372]
		Chronos-2	98	0.349	0.014	[0.323, 0.378]

- Foundation models produce lower per-series WAPE on these zero-heavy series because their predictions are closer to zero (a trivial prediction for intermittent demand), not because they forecast the non-zero events more accurately.

This tail behaviour is the mechanism behind the M5 WAPE artefact documented in §5 and is the reason we report an Excl-M5 sensitivity analysis in the generated replication artifacts. The M5 WRMSSE metric, which uses a hierarchical weighting scheme, does not suffer from the same denominator pathology; fair comparison requires computing it on a shared model/sample protocol.

Appendix D. Extraction Table (Abbreviated)

The extraction inputs (`analysis/extraction_schema.csv` and `benchmark/results/local_fm_sweep.csv`) record one row per (paper $\times$ model $\times$ dataset $\times$ metric) observation. The frozen extraction schema contains 802 rows; the local Source B sweep contributes 72 additional experiment rows. After canonicalising datasets, model families, and numeric metrics, the R pipeline retains 774 analysis-ready rows and forms $k = 498$ supporting Source A paired comparisons plus 45 Source B descriptive contrasts. The complete CSVs are available in the supplementary materials. Table D.17 shows the key columns; columns omitted for space include `dataset_variant`, `n_series`, `series_length_median`, `frequency`, `fine_tuned`, `runtime_reported`, `gpu_hours`, `hardware`, and `notes`. The R pipeline additionally derives `study_id`, `suite_id`, and `task_id`; the resulting study characteristics table is written to `analysis/study_characteristics.csv`. That file summarises analysis-contributing sources and benchmark suites after canonicalisation; it is not a one-row-per-PRISMA-record catalogue of all 32 included external source records.

The official fev-bench prediction-level rerun is stored separately from the frozen extraction table because it reruns benchmark windows rather than extracting published leaderboard rows. Its provenance is recorded in `analysis/figures/` through the official run manifest, model coverage, paired WAPE units, task contrasts, bootstrap draws, and covariance matrices. The generated file families follow the `fev_<model>_paired_wape_*.csv` pattern for Chronos-Bolt, Chronos-2, TiRex, and TimesFM 2.5.

Appendix D.1. Column Definitions

Appendix D.2. Row Counts by Category

Appendix D.3. Source A vs Source B

- **Source A** (literature extraction): categories A–F (757 schema rows). The structured-search flow counts 32 included external study/source records; the larger row count comes from model–dataset–metric rows and per-task suite re-extraction, not from 757 independent studies. These rows report accuracy numbers as published by the original authors or benchmark repositories.
- **Source B** (gap-filling experiments): OWN + LOCAL rows. The current model-level analysis uses the 72-row `local_fm_sweep.csv` export from our own experiments on consumer hardware (MacBook Air M3, 16 GB, MPS/CPU) and Azure ML (E4DS_V4), using the shared protocol of §5.1; earlier OWN rows are retained in the extraction schema for provenance.

Table D.17: Key columns in the extraction table.

Column	Description
`paper_id`	Unique row/result identifier, not a dependency cluster
`study_id`	Derived publication or experiment-source identifier
`suite_id`	Derived source-suite identifier (fev-bench, GIFT-Eval, Source B, other)
`task_id`	Derived benchmark task/dataset/frequency/horizon cell identifier
`authors`	First author et al.
`year`	Publication year
`dataset`	Target dataset (M5, Favorita, Rohlik, GIFT-Eval, fev-bench, etc.)
`model_name`	Model as named by the paper
`model_family`	One of: foundation, deep_learning, ml_tree, statistical, ensemble
`zero_shot`	Yes / No — whether the model was applied without task-specific training
`horizon`	Forecast horizon (days) or "mixed" if aggregated
`eval_method`	rolling / fixed_origin / competition
`metric_name`	Reported metric (WAPE, WRMSSE, MASE, MAE, sMAPE, WQL, CRPS, win_rate_*)
`metric_value`	Reported value (numeric or qualitative rank)
`has_covariates`	Whether the model used exogenous covariates

Table D.18: Extraction table row counts by category.

Category	Description	Rows
A	Foundation model papers	58
B	Benchmark papers and per-task re-extraction (incl. fev-bench, GIFT-Eval)	657
C	M5 competition and retail-specific	31
D	Cross-domain papers	3
F	Earlier fev-bench summary entries	8
OWN	Earlier Source B baseline rows in extraction schema	45
LOCAL	Local Source B sweep file (5 FMs + 3 baselines, MacBook/Azure)	72
Extraction schema		**802**
Local sweep file		**72**

The extraction table is frozen at the version used for the reanalysis of §6. Any post-submission additions would require re-running the generated contrast files and descriptive summaries of §6.

Appendix E. Threats to Validity

The analysis-contributing source assessment is generated as `analysis/risk_of_bias_assessment.csv`. It separates baseline quality, protocol matching, leakage risk, selective reporting, raw prediction availability, code availability, and independence notes for each `study_id`. Table E.19 summarises the resulting overall categories; Source B rows are not counted as included studies. This is a provenance and risk audit for the distinct sources that enter the analysis-ready extraction and benchmark reanalysis, not a full duplicate-screened risk-of-bias assessment for every manually screened candidate record or every one of the 32 Source A included studies. In Table E.19, the fev-bench count denotes six analysis-contributing `study_id` / source-slice records derived from the fev-bench extraction, not six separate fev-bench benchmark papers; the structured-search flow counts fev-bench itself as one benchmark source.

Table E.19: Risk/provenance assessment summary for analysis-contributing sources. This is not a duplicate-screened risk-of-bias assessment for all PRISMA-included studies.

Suite	Overall risk category	Sources
GIFT-Eval	moderate	1
SourceB	high (superseded unmatched runs) / moderate (matched panel)	1
fev-bench	moderate	6
other	high/unclear	3

Five threats specific to the Source B foundation model runs (§5.2), in decreasing order of magnitude:

1. **Extraction selection bias.** Papers that publish on M5 / Favorita / Rohlik are not a random sample of the FM literature — they are the subset that chose a retail task for their own evaluation. Papers that chose ETT / Weather / ECL (most of the LSF literature) are absent. The generated extraction and risk-of-bias artifacts surface this selection mechanism through source, suite, and provenance fields.

2. **Source A vs Source B protocol drift.** Source A numbers are extracted as-reported; Source B numbers are run on our §5.1 protocol. When the two diverge on the same (model, dataset, horizon) cell, the divergence is a data point about protocol sensitivity, not about model quality. The overlap between Source A and Source B is treated descriptively in §5.4, not as confirmatory paired evidence.

3. **Local run covers models up to 200 M params.** Moirai 2.0-Large (305 M) cannot be run locally (memory constraints, respectively) and is represented only by Source A. The five local models (Chronos-Bolt-Tiny 9 M, Moirai 2.0-Small 11 M, TiRex 35 M, Chronos-2 120 M, TimesFM 2.5 200 M) span over an order of magnitude in scale.

4. **Covariate asymmetry.** Most local FM families in Table 1 are univariate or are used in univariate zero-shot mode and cannot consume the §5.1 covariate sets. The LightGBM baselines in §5.3 use the full covariate sets. This biases the comparison toward LightGBM on covariate-driven datasets (M5: SNAP, prices; Favorita: oil, promotions) and is not correctable inside a zero-shot protocol. The moderator discussion in §6.2 treats covariate-aware vs univariate comparisons as hypothesis-generating.

5. **Top-30k Favorita cap propagates to the local run.** Same cap, same selection rule, same velocity bias as §5.1. Source A rows for Favorita include both fev-bench's top-54k subset and the full-Favorita subset where available; we flag `dataset_variant` per row so the §6 pool can restrict to comparable subsets.

Appendix F. Detailed Baseline Reliability Analysis

This appendix keeps the per-dataset baseline tables used by §5.3. The main analysis treats the direct-vs-recursive pattern as a fixed-budget diagnostic, not as a causal intermittency result. Three reproducibility checks matter most: (i) the earlier fixed-budget protocol ranking differs across datasets, (ii) the matched-panel aggregate-WAPE view makes M5 and Rohlik more conservative than the earlier per-series view and gives a mixed Favorita protocol ranking, and (iii) the 200-trial M5 LGBM sensitivity changes the tuning-budget interpretation without removing the Chronos-2 M5 advantage. The Favorita `direct_scaled` run should not be used as inferential evidence.

Appendix F.1. M5 Results

Table F.20 reports the rolling-origin M5 diagnostics. WRMSSE ranks direct LightGBM ahead of recursive LightGBM and Seasonal Naive within this protocol. These rows are not compared with the official M5 competition winner because the competition used a fixed 28-day task and a different evaluation setup.

Table F.20: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the M5 tail-eval window. Bold = best per horizon.

Model	h	MAE	sMAPE	WAPE	WRMSSE	Runtime (s)	Cost (USD)
Seasonal Naive	7	1.1590	75.44	1.3072	0.7758	236.9	0.025
Seasonal Naive	14	1.1783	76.06	1.3216	0.7718	176.9	0.019
Seasonal Naive	28	1.1897	76.69	1.3285	0.7661	139.6	0.015
LightGBM rec	7	1.0110	144.44	1.6246	0.6336	300.3	0.032
LightGBM rec	14	1.0391	144.55	1.7292	0.6639	301.1	0.032
LightGBM rec	28	1.0743	144.48	1.9357	0.6976	284.3	0.030
LightGBM dir	7	**0.9682**	145.81	**1.3512**	**0.5598**	1114.4	0.118
LightGBM dir	14	**0.9896**	146.20	**1.4100**	**0.5821**	2003.2	0.212
LightGBM dir	28	**1.0117**	146.95	**1.5133**	**0.5992**	3728.8	0.394

On M5, direct LightGBM is better than recursive LightGBM on both WAPE and WRMSSE in the fixed-budget sweep. Seasonal Naive has lower per-series WAPE than direct LightGBM at every horizon, despite worse WRMSSE, which is why M5 WAPE is treated as a denominator-pathology warning rather than as the sole ranking metric.

Appendix F.2. Rohlik Results

Table F.21: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Rohlik v2 tail-eval window. Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	40.649	36.46	0.4015	8.5	0.0009
Seasonal Naive	14	42.236	37.20	0.4031	6.9	0.0007
Seasonal Naive	28	43.543	38.21	0.4116	5.8	0.0006
LightGBM rec	7	**19.335**	92.90	**0.3654**	29.6	0.0031
LightGBM rec	14	20.972	94.59	**0.3953**	30.6	0.0032
LightGBM rec	28	22.815	97.18	**0.4321**	29.1	0.0031
LightGBM dir	7	19.549	93.34	0.3816	141.1	0.0149
LightGBM dir	14	**20.210**	94.17	0.4072	265.6	0.0280
LightGBM dir	28	**21.087**	95.01	0.4442	524.3	0.0553

Rohlik reverses the M5 fixed-budget WAPE ranking: recursive LightGBM is lower than direct at all three horizons, while direct is several times more expensive. This is a protocol-and-dataset interaction hypothesis, not a standalone intermittency proof.

Appendix F.3. Favorita Results And `direct_scaled` Flag

Table F.22: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Favorita top-30k tail-eval window. Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	5.248	60.76	0.6363	65	0.0068
Seasonal Naive	14	5.400	61.49	0.6473	49	0.0052
Seasonal Naive	28	5.644	62.49	0.6643	39	0.0041
LightGBM rec	7	**2.507**	90.92	**0.5295**	236	0.0249
LightGBM rec	14	**2.574**	91.32	**0.5311**	230	0.0242
LightGBM rec	28	2.692	92.13	**0.5377**	230	0.0243
LightGBM dir	7	2.533	91.38	0.5513	1,015	0.1071
LightGBM dir	14	2.619	92.17	0.5711	1,907	0.2012
LightGBM dir	28	**2.688**	92.75	0.5892	3,588	0.3787

In the fixed-budget Favorita sweep, recursive LightGBM is cheaper and lower on WAPE than direct LightGBM. A follow-up `direct_scaled_favorita.csv` run is marked as superseded because its $h = 7$ row changes WAPE despite the same nominal 300-tree count as the base direct run. The matched-panel protocol check below replaces that run for interpretation.

Appendix F.4. Descriptive Source B Grid

The grid below reports per-series WAPE means from the saved unmatched Source B result file. Because `n_series` varies within every dataset–horizon cell, the table is provenance material and should not be read as a fully paired model comparison. The per-series denominator makes M5's tail-sparse rows sensitive to intermittent-demand artifacts.

Table F.23: Source B WAPE grid (per-series mean, rolling origin) across models, datasets, and horizons. Lower is better. The grid is complete, but `n_series` differs across models; see `source_b_n_series_matrix.csv`.

Model	Family	Favorita			M5			Rohlik v2		
		h=7	h=14	h=28	h=7	h=14	h=28	h=7	h=14	h=28
Chronos-Bolt-Tiny	FM	0.482	0.485	0.489	0.897	0.902	0.911	0.334	0.345	0.361
Moirai 2.0-Small	FM	0.477	0.482	0.487	0.887	0.893	0.901	0.324	0.337	0.354
TiRex	FM	0.525	0.531	0.539	0.903	0.907	0.914	0.323	0.337	0.353
Chronos-2	FM	0.477	0.481	0.486	0.879	0.887	0.899	0.324	0.339	0.357
TimesFM 2.5	FM	0.528	0.534	0.542	0.944	0.944	0.948	0.324	0.338	0.353
`lightgbm_cov`	ML_TREE	0.530	0.531	0.538	1.625	1.729	1.936	0.365	0.395	0.432
`lightgbm_direct`	ML_TREE	0.551	0.571	0.589	1.351	1.410	1.513	0.382	0.407	0.444
`seasonal_naive`	STATS	0.636	0.647	0.664	1.307	1.322	1.329	0.402	0.403	0.412

Table F.24: Earlier full-workload summary of the fixed-budget direct-vs-recursive LightGBM gap using per-series WAPE means (positive = direct lower, negative = recursive lower).

Dataset	Series-day zero frac.	$h = 7$	$h = 14$	$h = 28$	Interpretation
M5	∼70 %	+16.8 %	+18.5 %	+21.8 %	Direct lower in this protocol
Rohlik	∼1.2 %	−4.4 %	−3.0 %	−2.8 %	Recursive lower and cheaper
Favorita	∼15–25 %	−4.1 %	−7.5 %	−9.6 %	Recursive lower in fixed-budget run

Appendix F.5. Earlier Fixed-Budget Protocol Summary

Appendix F.6. Matched-Panel Protocol Check

The matched-panel analysis recomputes the M5, Rohlik, and Favorita LightGBM protocol contrast on the same 100 top-volume series used for the FM-vs-baseline paired panel. Using aggregate WAPE, direct is slightly worse than recursive on M5 at $h = 7$ and $h = 14$, slightly better at $h = 28$, and worse than recursive on Rohlik at all horizons. On Favorita, recursive is lower at $h = 7$ and $h = 14$, while direct is lower at $h = 28$. The horizon-scaled direct run equals the unscaled direct run at $h = 7$ on all three datasets, as expected when both use 300 trees. This resolves the same-tree-count behaviour inside the new matched-panel protocol; it does not retroactively explain the earlier unmatched `direct_scaled_favorita.csv` anomaly.

The combined evidence supports a cautious hypothesis: the best LightGBM protocol can change with the dataset, metric aggregation, sampled workload, and training budget. It does not support a hard intermittency threshold, nor a general rule that direct or recursive is always preferable.

Appendix F.7. Cost Consistency Audit

The Azure cost tables are retained for audit, but this paper does not use them for universal cost conclusions. The detail tables and summary table do not yet reconcile to one ledger.

The cost plots in §6.4 therefore remain cost-error scatterplots. The matched-panel analysis writes a local run ledger for the completed M5, Rohlik, and Favorita reruns. The older Azure summary/detail tables still need a single reconciled ledger before cost claims are promoted beyond descriptive accounting.

Table F.25: Baseline cost consistency audit. Differences are too large to treat as rounding; a single run ledger is required before cost claims are promoted from descriptive to inferential.

Scope	Detail tables (USD)	Summary table (USD)	Difference	Status
M5 total	0.877	0.810	−0.067	mismatch requires single ledger
Rohlik total	0.110	0.120	+0.010	mismatch requires single ledger
Favorita total	0.777	1.119	+0.343	mismatch requires single ledger
Favorita direct	0.687	1.030	+0.343	mismatch requires single ledger
M5 seasonal naive	0.059	0.146	+0.087	mismatch requires single ledger
M5 direct	0.724	0.571	−0.153	mismatch requires single ledger

Appendix F.8. Reproducibility Detail

Every baseline result should be tied to a git commit SHA, an Azure ML pipeline run name, a registered data asset version, and a versioned environment. The matched-panel runs store shared series identifiers, forecast origins, actuals, per-model predictions, and paired bootstrap contrasts for completed M5, Rohlik, and Favorita cells. The same requirement applies to future GPU/larger-workload reruns and covariate-aware FM variants before Source B can enter a broader formal paired analysis.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Maciej Rubczynski: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

A historical technical-report snapshot is archived on Zenodo at Zenodo. The live repository for the current analysis is available at GitHub; the release package corresponding to this manuscript is prepared as a versioned Zenodo-ready archive under `release/` rather than overwriting the historical snapshot, and is prepared as GitHub release v1.21 at the v1.21 release. The package records the source commit in `COMMIT.txt` and file hashes in `CHECKSUMS.txt`. The repository includes derived extraction tables, analysis code, generated tables/figures, study-characteristics data, audit outputs, and replication scripts, but not raw competition data or untracked fev-bench prediction parquet trees. The M5 dataset is publicly available via Kaggle; the Favorita dataset is publicly available via Kaggle; the Rohlik v2 dataset is available via the Rohlik v2 Kaggle competition.

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Declaration of generative AI in scientific writing

During manuscript preparation, the author used OpenAI Codex to assist with language editing, document organization, and code checks. The author reviewed and edited all outputs and takes full responsibility for the content of the article.

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